

BSc

COMPUTER SCIENCE



Reinvent the Future

The US Bureau of Labour Statistics projects a 24% growth for Computer Science jobs through 2026. This statistic represents significant opportunities for Computer Science students as they prepare for the world of work. Computer science programs around the world continue to evolve, driven by technological advancements and industrial requirements. They constitute the creative minds behind computer programs. Many computer scientists working across multiple geographic regions collaborate to develop applications that allow people to do specific tasks on a computer or another device. Some are involved in the development of the underlying systems that run the devices or that control ubiquitous networks across the globe.

Computer Science at ACity

The Computer Science program aims to develop professionals, who can apply broad knowledge in mathematics, science, and technology to analyze complex problems, design, implement and evaluate computing-based solutions in industries, academia, and government. They would also be able to expand human knowledge in evolving fields in computational sciences (blockchains, AI, Robotics, IoT, etc.) through research and development and to establish a community of experts to serve as the nucleus for industrialization and national development.

This curriculum delineates pathways for students hoping to pursue advanced studies in computer science by ensuring that specific electives allow them to focus on specific areas of concentration.

Top Careers in Computer Science

Some possible career options with a degree in Information Technology include:

- 👉 Information Systems Manager
- 👉 Machine Learning Engineer
- 👉 Multimedia Programmer
- 👉 Database Administrator
- 👉 Applications Developer
- 👉 Cybersecurity Analyst

Why ACity?

Home to
students from over
23 countries
across Africa

A fully digitized
campus with the fastest
internet speed
in a Ghanaian university

70%
practical lessons

**Experienced
Faculty**
with Industry Expertise

Every **ACitizen** has
the opportunity to do a
work placement
every vacation

Student-To-Faculty Ratio
5:1

Houses a
Digital Makerspace
and a **Technology &
Entrepreneurial Center**
for **Creative freedom
& Innovation**

Our Unique Learning Pillars



Experiential Learning

Hands-on learning to prepare students to readily apply concepts, to easily integrate into the workspace.



Contextual Learning

Solving real grass-root problems to expose students to the local context and develop empathy towards the continent's progress.



Unified Learning

A project-based approach that combines concepts across courses to connect the dots and enable unified learning.



Extensional Learning

Arms students with a viable toolkit to help them confront real-life issues, they may not have encountered during their academic life, squarely.

Entry Requirements

WASSCE/NECO

Minimum C6 in 6 subjects including 3 core subjects (Maths and English mandatory) and 3 elective subjects

SSCE

Minimum D or a pass in 6 subjects including 3 core subjects (Maths and English mandatory) and 3 elective subjects

IGCSE O-LEVEL & A-LEVEL

Minimum of 5 credit passes in the IGCSE/O-Levels (Maths and English mandatory) and 3 passes in the A-Levels

ENGLISH IB

Minimum of 5 credit passes in the IGCSE/O-Levels (Maths and English mandatory) and a minimum score of 4 points in 3 Higher Level (HL) subjects

FRENCH IB

Minimum of 50% overall average pass. (Results to be presented to NAB for verification) Maths and English mandatory

AMERICAN HIGH SCHOOL

Minimum GPA of 3.0 (Maths and English mandatory)

COURSE OUTLINE

SEMESTER 1

Course Name

Communication Skills
Data Analysis (using MS Excel)
French Language
Fundamentals of Innovation and Entrepreneurship (FIE) Seminar I
Introduction to Programming with Python
Physical Sciences
Pre-Calculus (with MATLAB)
Technology and Society

SEMESTER 2

Course Name

Analytic Geometry and Calculus I (with MATLAB)
Emerging Technologies
Financial Accounting I
Fundamentals of Innovation and Entrepreneurship (FIE) Seminar II
Introduction to Electrical and Electronics Engineering
Logic and Critical Thinking
Programming in C
Text and Meaning

SEMESTER 3

Course Name

Analytic Geometry and Calculus II (with MATLAB)
Data Structures and Algorithms
Fundamentals of Innovation and Entrepreneurship (FIE) I
Fundamentals of Logic Design
Introduction to Operations Research
Leadership Seminar I
Object Oriented Programming with C++

SEMESTER 4

Course Name

African Studies
Applied Linear Algebra (with MATLAB)
Computer Architecture and Organisation
Design and Analysis of Algorithms
Fundamentals of Innovation and Entrepreneurship (FIE) II
Probability, Statistics and Reliability (with MATLAB)
Programming with Java

SEMESTER 5

Course Name

Computer Networks
Database Systems
Discrete Mathematics (with MATLAB)
Leadership Seminar II
Numerical Methods (with MATLAB)
Object Oriented System Design
Operating Systems
Web Technologies

SEMESTER 6

Course Name

Computer Graphics and Multimedia
Distributed Computing
Industry Internship
Principles of Compiler Design
Project Management, Engineering Economics and Risk Analysis
Research Methods in Computing
Software Engineering

SEMESTER 7

Course Name

Computer Security

Computer Systems Programming

Human-Computer Interaction

Project Phase I

Technical Elective - I

Technical Elective - II

SEMESTER 8

Course Name

Introduction to Artificial Intelligence

Mobile Computing

Professional Ethics and Values

Project Phase II

Technical Elective - III

Technical Elective - IV

ELECTIVES

Data Processing and Networks

Cryptography and Network Security
Neural Networks
Data Mining and Warehousing
TCP/IP Design and Implementation

Modern Computing and Programming

E-commerce
Programming in ASP.NET
Advanced Java Programming
Cloud Computing

Advanced and Emerging Technologies

Embedded Systems
Introduction to Robotics
Internet of Things (IoT)
Machine Learning

Computational Sciences and Data Analytics

Business Intelligence
Cloud Application
Introduction to Data Science
Machine Learning